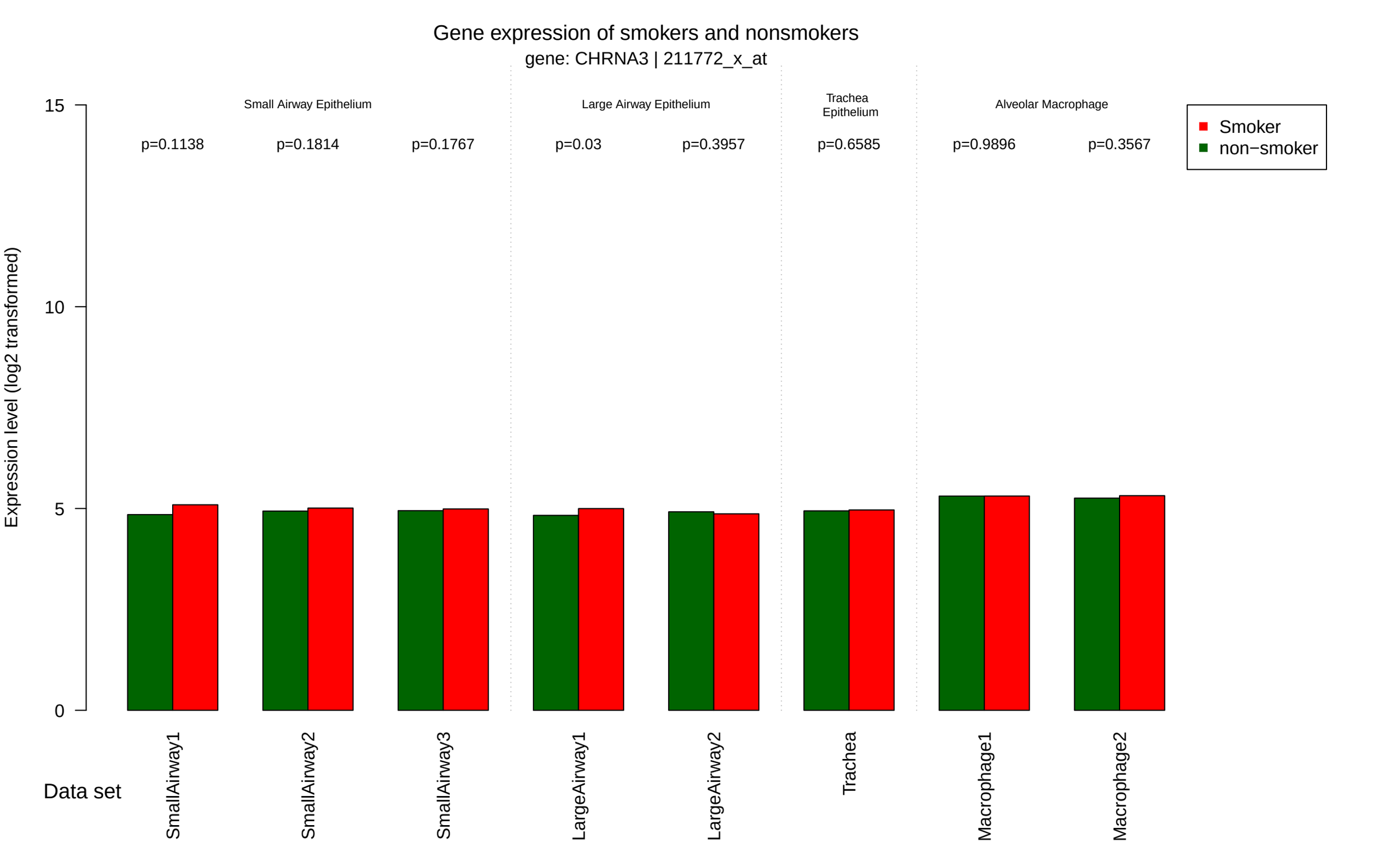
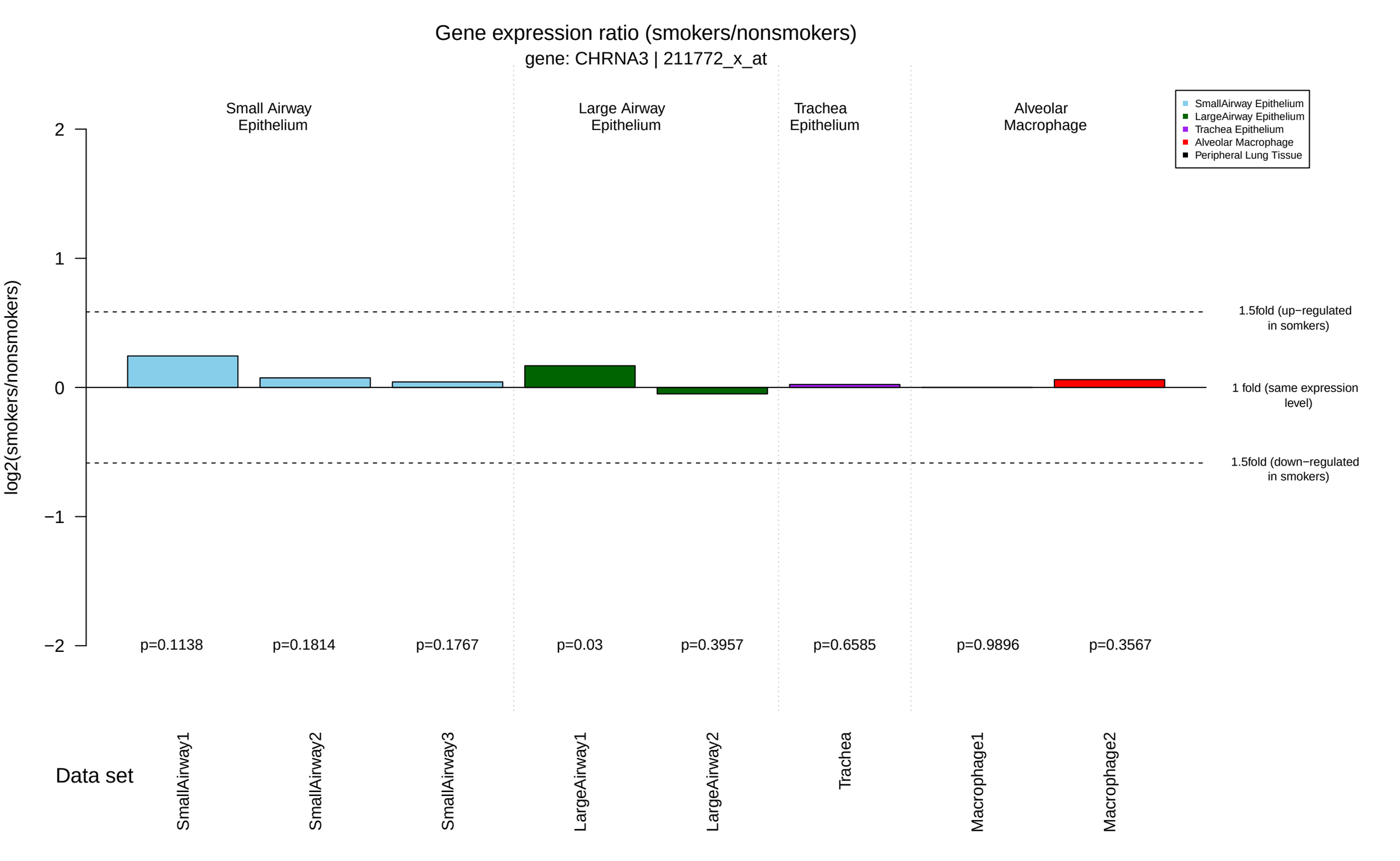
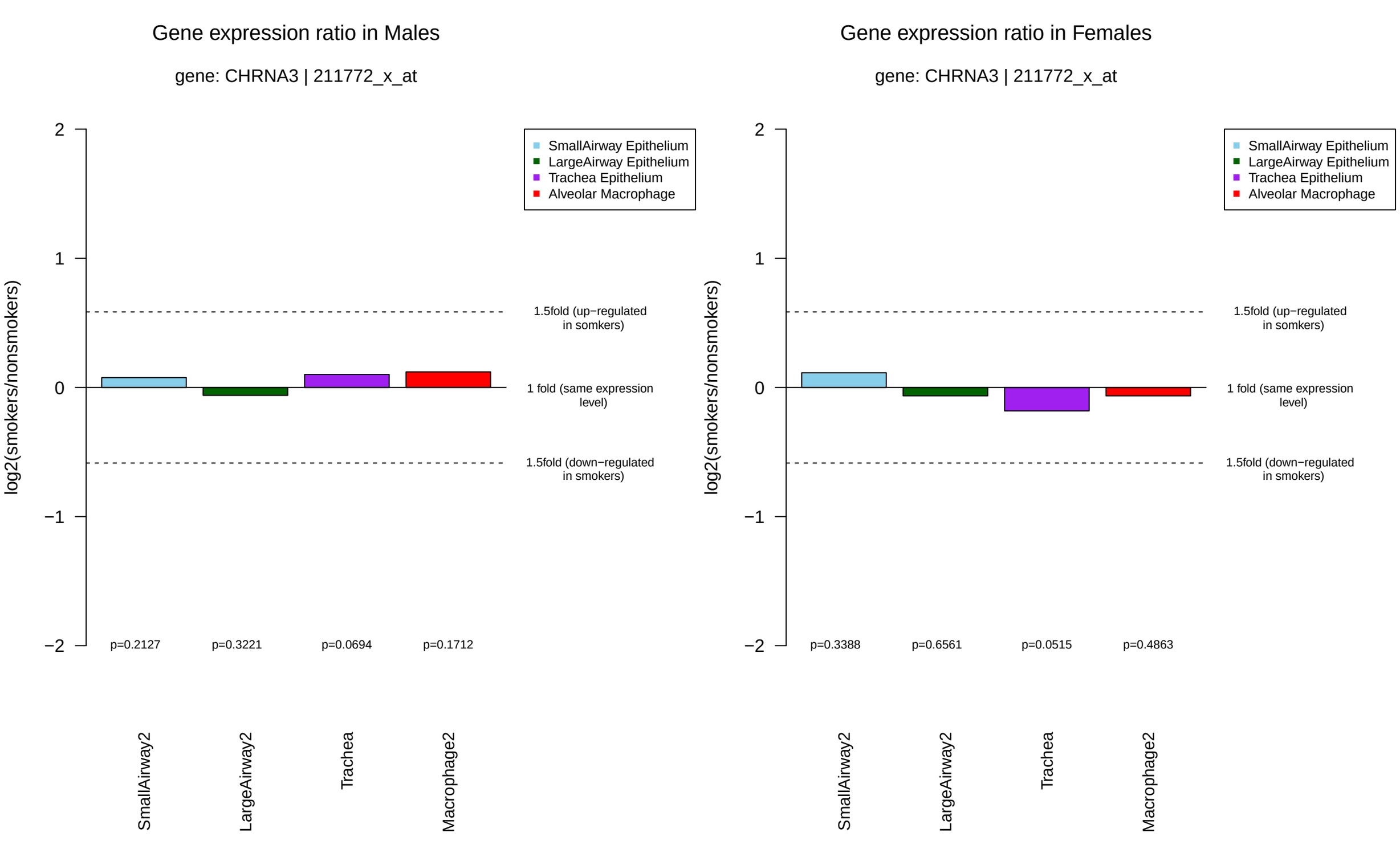
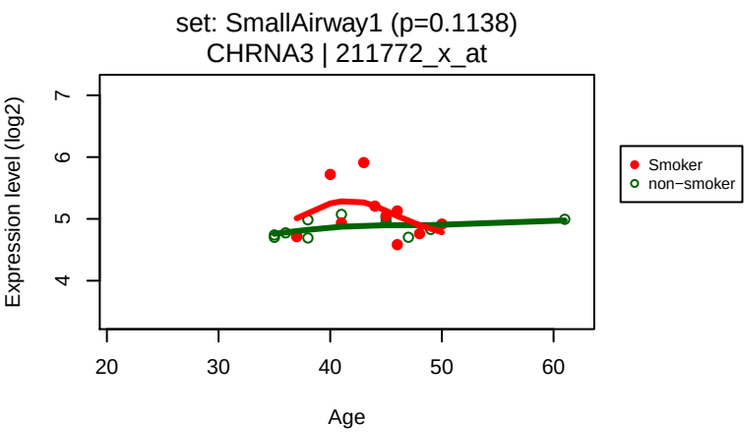
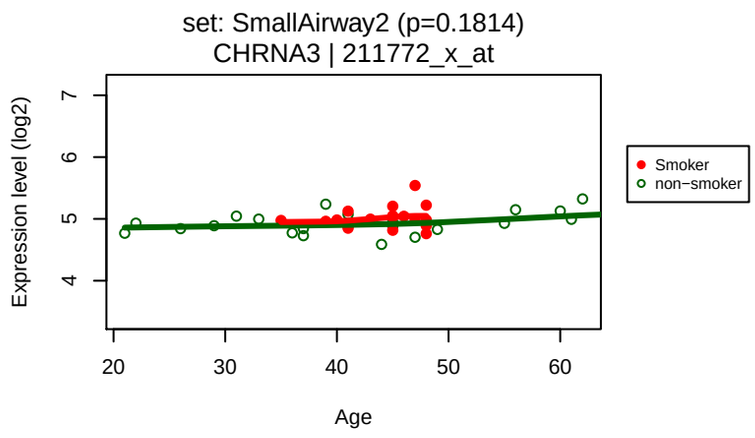
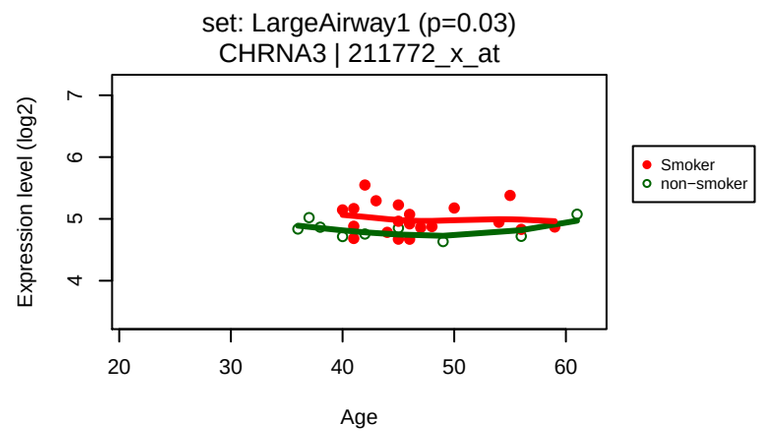

Gene expression of smokers and nonsmokers
gene: CHRNA3 | 211772_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1138
p=0.1814
p=0.1767
p=0.03
p=0.3957
p=0.6585
p=0.9896
p=0.3567
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

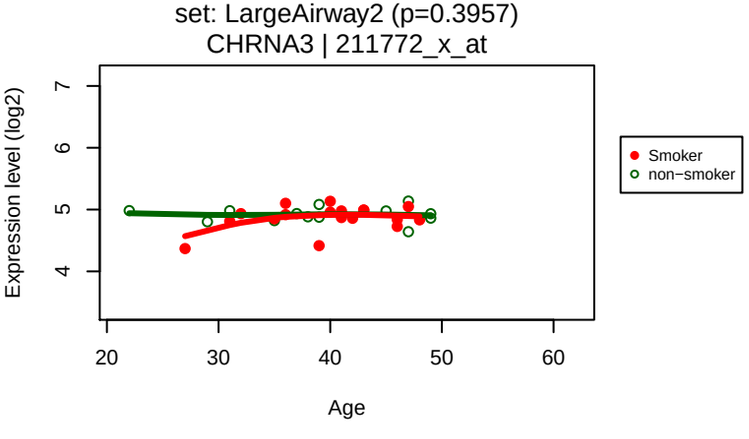

Gene expression ratio (smokers/nonsmokers)
gene: CHRNA3 | 211772_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1138
p=0.1814
p=0.1767
p=0.03
p=0.3957
p=0.6585
p=0.9896
p=0.3567
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

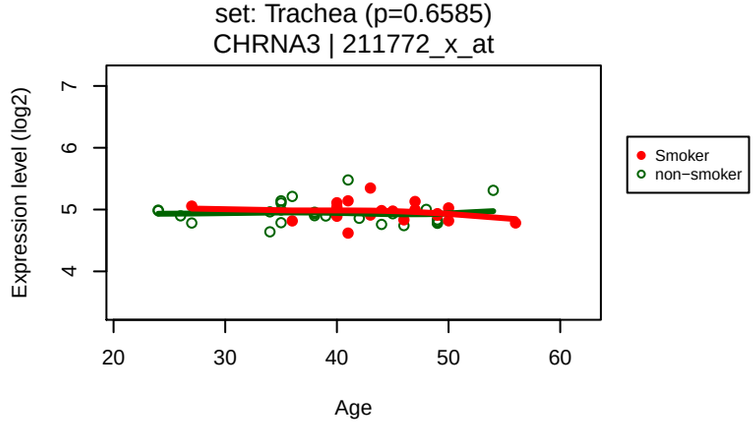

Gene expression ratio in Males
gene: CHRNA3 | 211772_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2127
p=0.3221
p=0.0694
p=0.1712
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRNA3 | 211772_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3388
p=0.6561
p=0.0515
p=0.4863
SmallAirway2
LargeAirway2
Trachea
Macrophage2

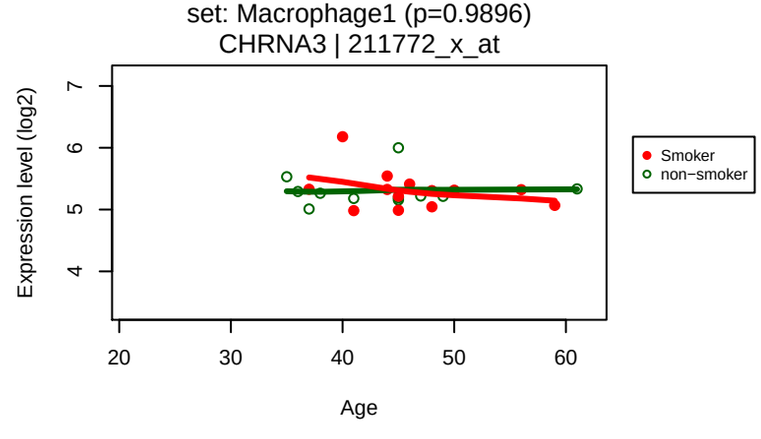

set: SmallAirway1 (p=0.1138)
CHRNA3 | 211772_x_at
Expression level (log2)
Age
Smoker
non−smoker

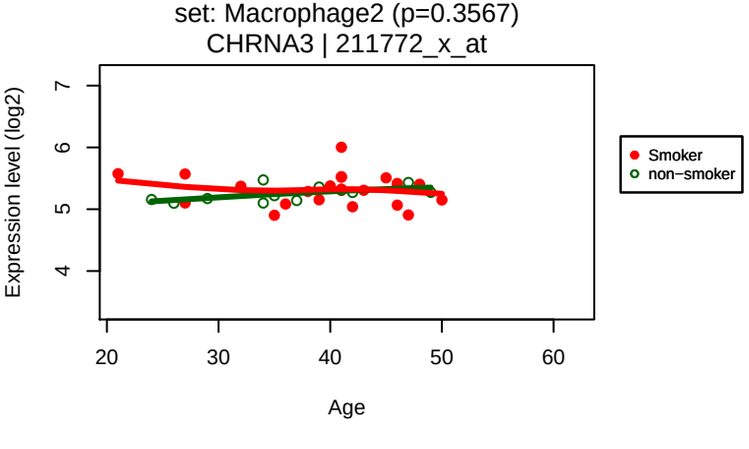

set: SmallAirway2 (p=0.1814)
CHRNA3 | 211772_x_at
Expression level (log2)
Age
Smoker
non−smoker

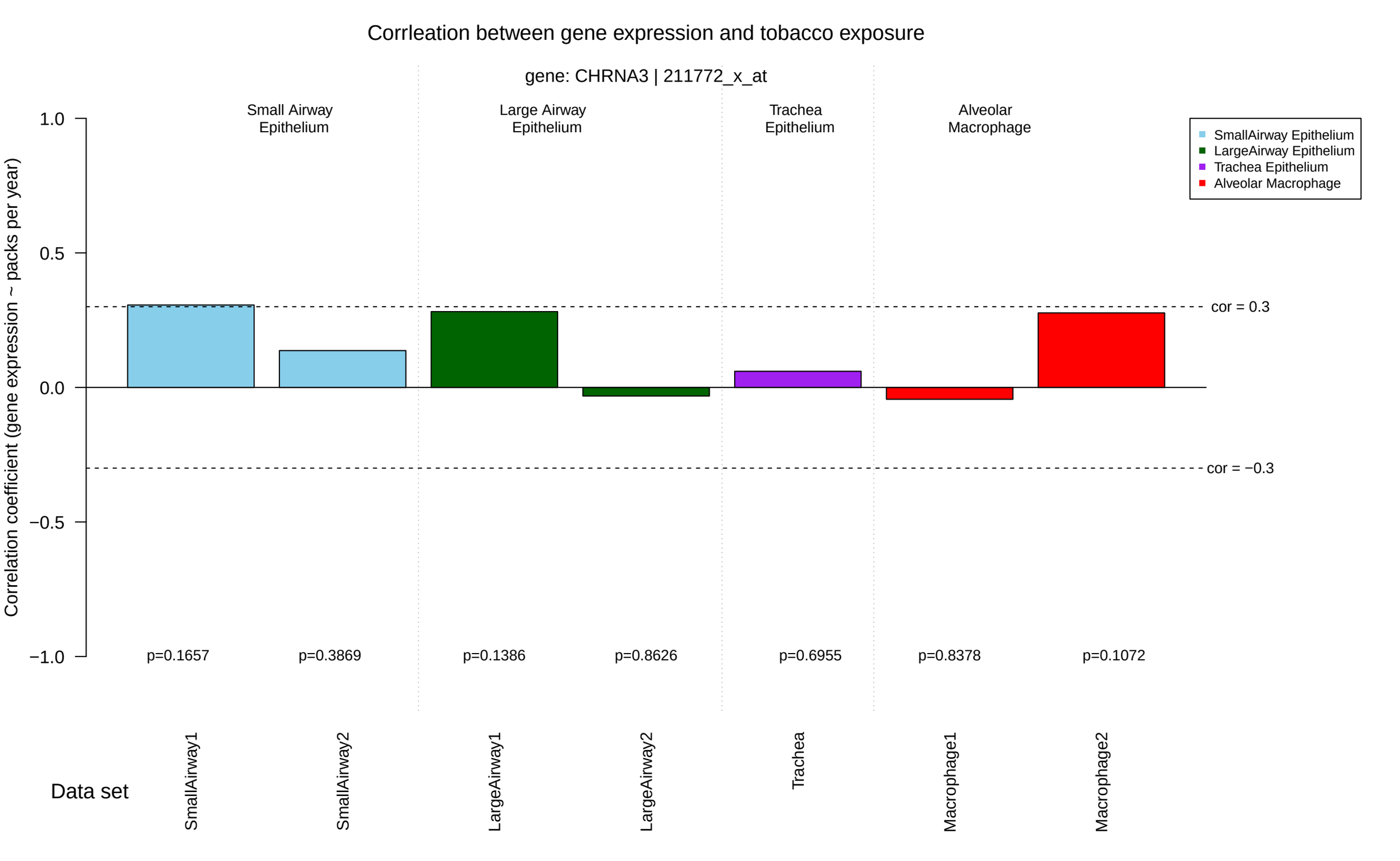

set: LargeAirway1 (p=0.03)
CHRNA3 | 211772_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3957)
CHRNA3 | 211772_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6585)
CHRNA3 | 211772_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9896)
CHRNA3 | 211772_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3567)
CHRNA3 | 211772_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRNA3 | 211772_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1657
p=0.3869
p=0.1386
p=0.8626
p=0.6955
p=0.8378
p=0.1072
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2